

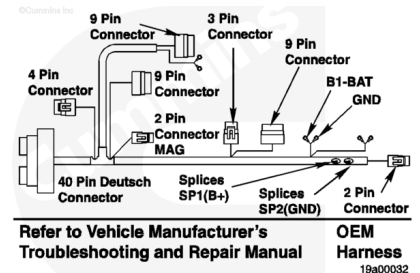
Trainings ▾

General Information

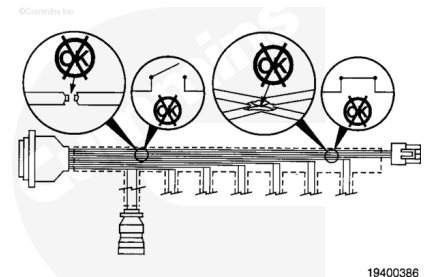
The QST30 Power Generation Interface engine uses extension cables and an adapter cable to attach the engine harness to the two CM552 engine control modules (ECMs) and the one CM850 ECM.

The engine also uses two separate wiring harnesses to control the engine and some of the equipment operations:

- Engine harness.
- Original equipment manufacturer (OEM) harness.



Replace a harness or extension/adapter cable, if an open circuit or a short circuit is found under the protective covering of the harness body.

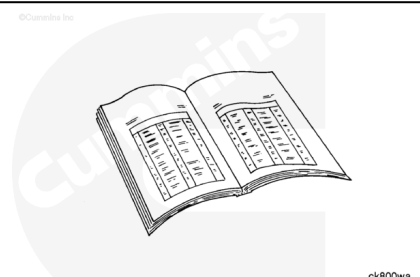


Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. Refer to equipment manufacturer service information.



Remove

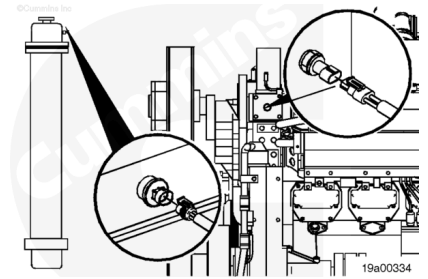
Disconnect the engine harness from the electrical system at the INLINE™ E connector.

Disconnect the engine harness from the coolant pressure sensor, if equipped.

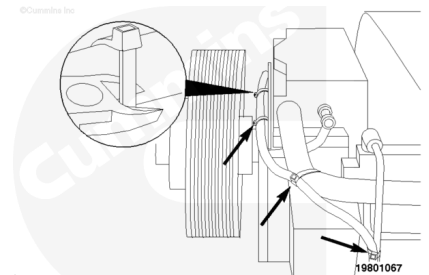
Remove the engine harness clamps.

Disconnect the engine harness from the coolant temperature sensor and coolant level sensors.

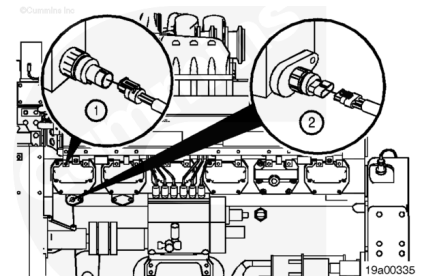
Disconnect the engine harness from the coolant level harness extension connector.



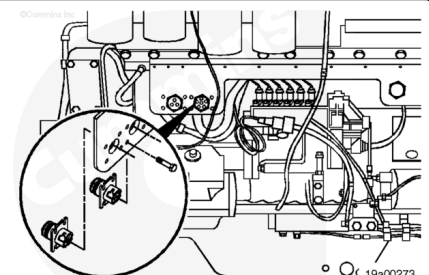
Cut the cable ties that secure the coolant level sensor extension harness to the thermostat housing bracket, aftercooler tube, and fuel tube.



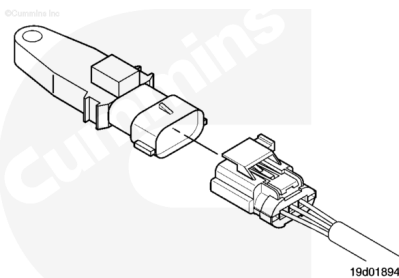
Disconnect the engine harness from the left bank intake manifold pressure sensor (1 or 2, depending on mounting location).



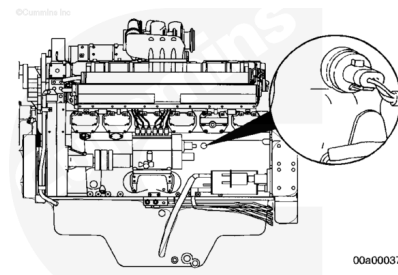
Disconnect the two 6-pin or one 9-pin data link connector(s) from the harness support bracket by removing the capscrews.



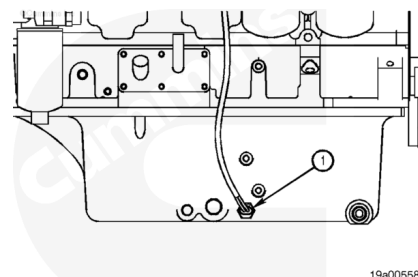
Disconnect the INLINE™ connector bracket, which includes the barometric pressure sensor connector.



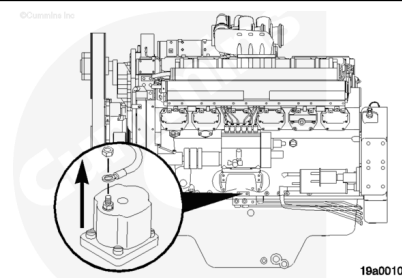
Disconnect the engine harness from the oil pressure sensor.



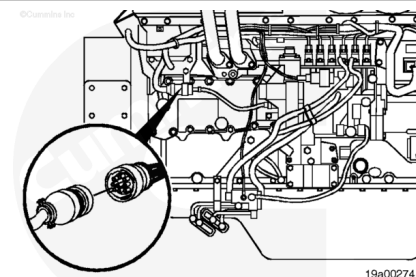
Disconnect the engine harness from the oil temperature sensor, if equipped.



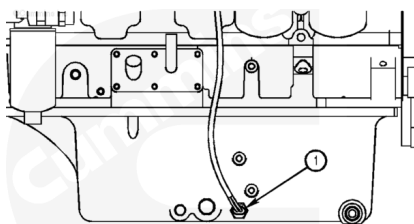
Disconnect the engine harness fuel shutoff control wires from the fuel shutoff solenoids (left bank and right bank).



Disconnect the engine harness from the fuel pump connectors (left bank and right bank).

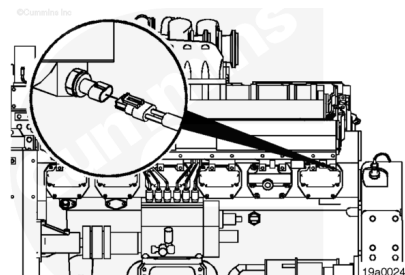


Disconnect the engine harness from the Centinel™ oil level extension harness, if equipped.



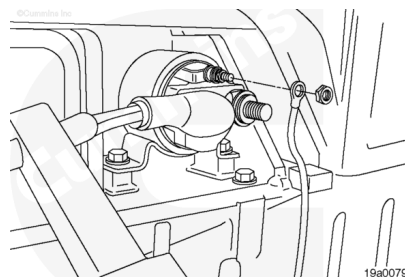
19a00558

Disconnect the engine harness from the left bank intake manifold 1 temperature sensor.



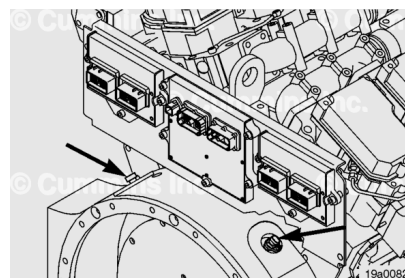
19a00247

Disconnect the engine harness from the grid heater relay connector, if equipped.



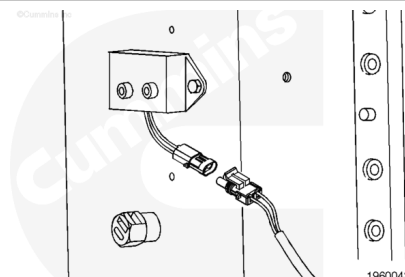
19a00793

Disconnect the engine harness from the engine speed sensor.



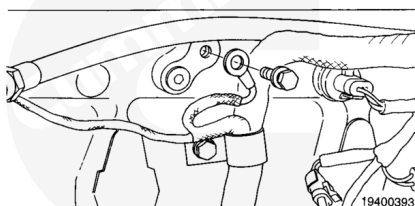
19a00829

Disconnect the engine harness from the Centinel™ tank switch extension, if equipped.

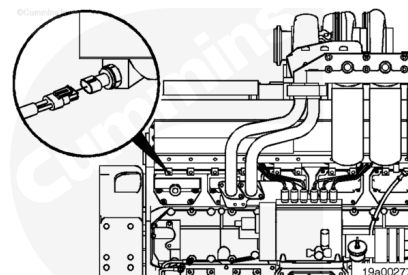


19600430

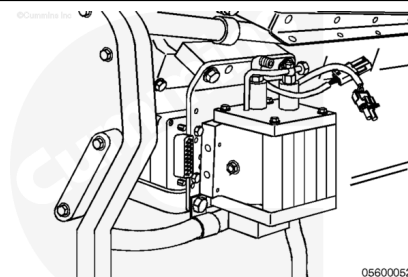
Disconnect the engine harness engine block ground from the engine block.



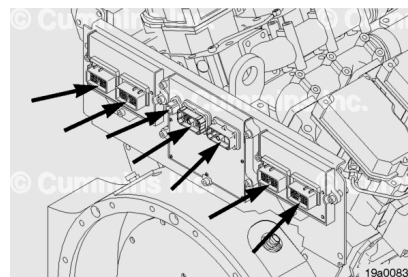
Disconnect the engine harness from the right bank intake manifold 2 temperature sensor connector.



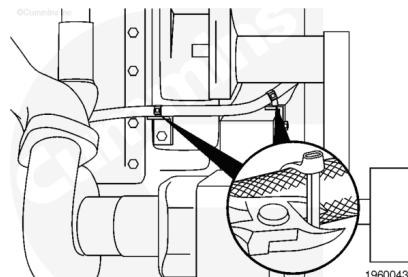
Disconnect the engine harness from the Centinel™ actuator connector, if equipped.



Use a 4 mm [5/32 in] hex head wrench to disconnect the engine harness Deutsch™ connectors from the two CM552 ECMs and the one CM850 ECM.

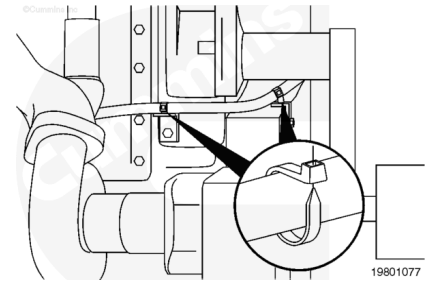


Cut the remaining ties holding the engine harness to the engine. Remove the engine harness.



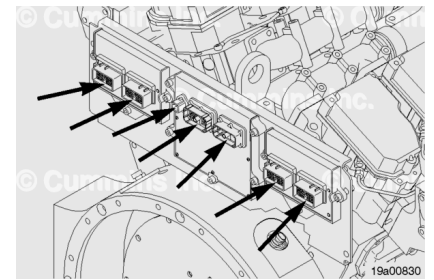
Install

Connect the engine harness to the engine using cable ties.



Connect the engine harness Deutsch™ connector to the engine harness extension cable, if used, or the engine harness adapter cable.

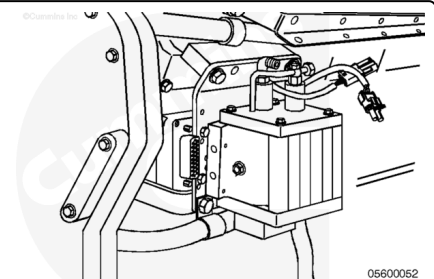
Use an inch-pound torque wrench, Part Number 3376592, with 4 mm [5.32 in] hex head adapter to tighten the connector jackscrew.



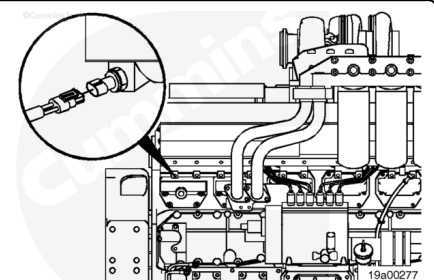
Torque Value: 2.8 n•m [25 in-lb]

Note : Do not over torque as connector damage can occur.

Connect the engine harness to the Centinel™ actuator connector, if equipped.

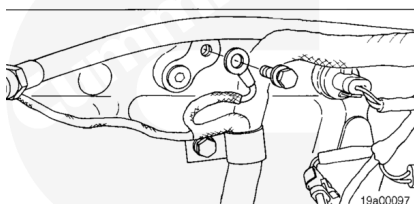


Connect the engine harness to the right bank intake manifold 2 temperature sensor connector.



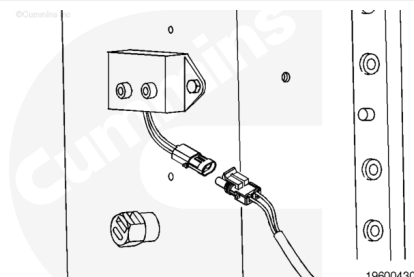
Connect the engine block ground to the engine block.

Torque Value: 20 n•m [177 in-lb]



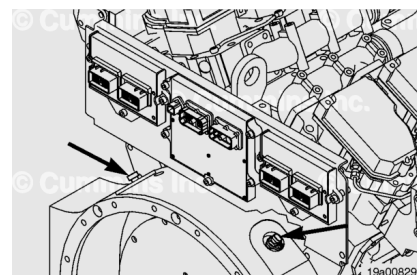
19a00097

Connect the engine harness to the Centinel™ tank switch extension, if equipped.



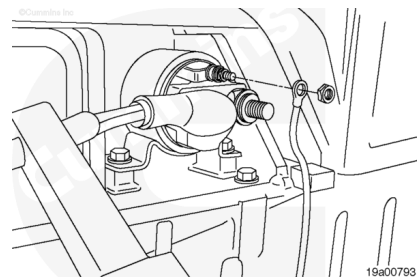
19600430

Connect the engine harness to the engine speed sensors.



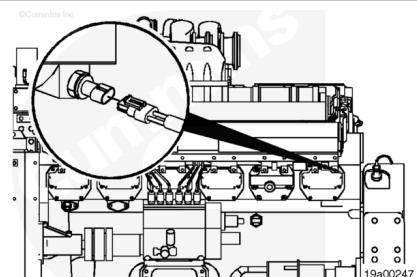
19a00829

Connect the engine harness to the grid heater relay connector, if equipped.



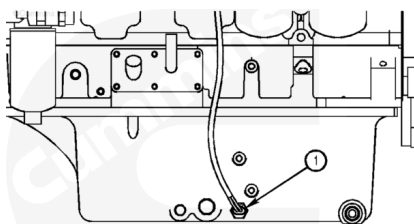
19a00793

Connect the engine harness to the left bank intake manifold 1 temperature sensor.



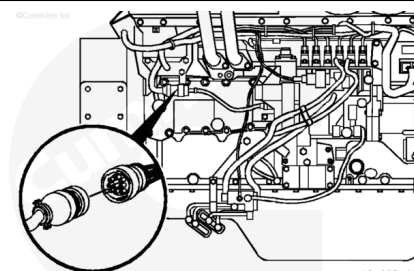
19a00247

Connect the engine harness to the Centinel™ oil level extension harness, if equipped.



19a00558

Connect the engine harness to the fuel pump connectors (left bank and right bank).

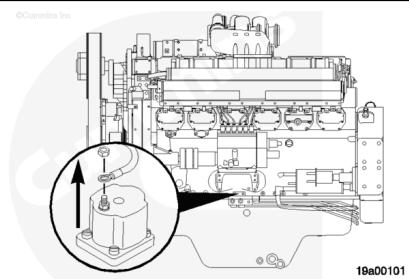


19a00274

Connect the fuel shutoff control wires to the fuel shutoff solenoids, (left bank and right bank).

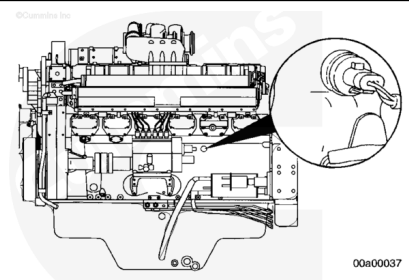
Tighten the retaining nut.

Torque Value: 3 n·m [27 in-lb]



19a00101

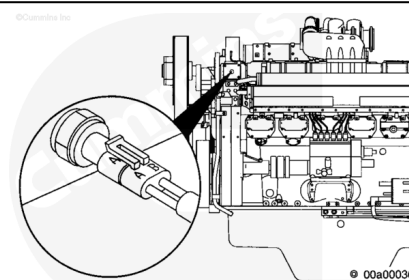
Connect the engine harness to the oil pressure sensor.



00a00037

Connect the engine to the coolant temperature sensor.

Connect the engine harness clamps.



00a00036

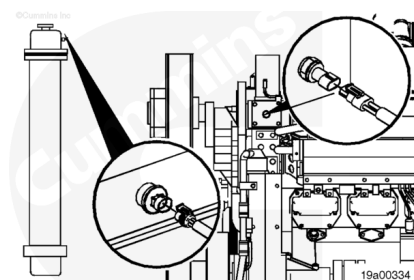
Connect the engine harness to the electrical system at the INLINE™ E connector.

Connect the engine harness extension cable to the extension cable connector.

Connect the engine harness adapter cable and all engine harness extension cables used.

Use a 4mm [5/32 in] hex head wrench to tighten the capscrews.

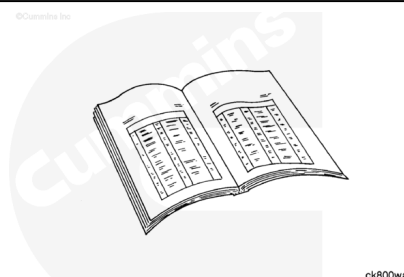
Torque Value: 3 n·m [27 in-lb]



Finishing Steps

⚠ WARNING ⚠

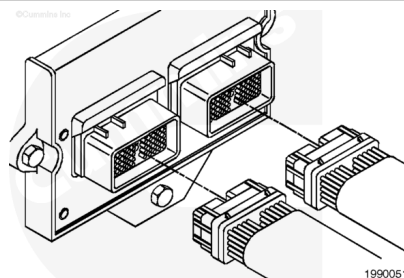
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Connect the batteries. Refer to the OEM service manual.

Resistance Check

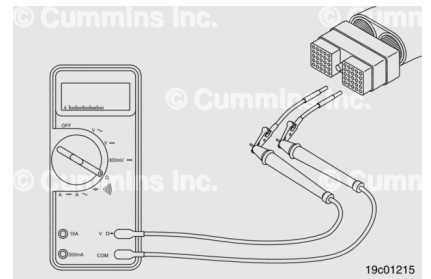
Disconnect the right bank and left bank ECM connectors.



Left Bank

⚠ CAUTION ⚠

Do not use probes or test leads other than Part Number 3822758. The connector will be damaged. The leads must fit tightly in the connector without expanding the pins in the connector.



Measure the resistance from the rack position common wire to the battery 1 voltage return on the ECM 1 CM552 (Source Address 00) 50-pin connector. Refer to the wiring diagram for connector pin identification. No resistor is used on engine wiring harness, Part Number 4975505.

The resistance **must** be between 1261 and 1339 ohms.

Note : The resistance value must be between 2134 and 2266 ohms for engine wiring harness, Part Numbers 2881121 and 4975747.

If the harness resistance value between the harness connector pins is within the specification, consult the fault code or symptom based troubleshooting procedures.

If the harness resistance value does **not** meet the specification, replace the resistor and check the resistance value again.

If the harness resistance still does **not** meet specification, replace the harness.

Right Bank

Measure the resistance from the rack position common wire to the battery 1 voltage return on ECM 3 CM552 (Source Address 01) 50-pin connector. Refer to the wiring diagram for connector pin identification.

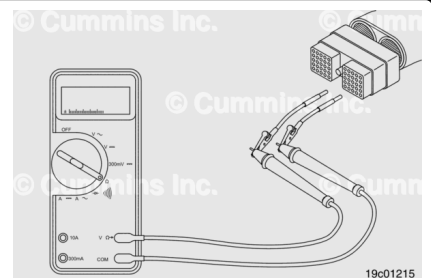
The resistance **must** be between 1261 and 1339 ohms.

Note : The resistance value must be between 2134 and 2266 ohms for engine wiring harness, Part Numbers 4975505, 2881121, and 4975747.

If the harness resistance value between the harness connector pins is within the specification, consult the fault code or symptom based troubleshooting procedures.

If the harness resistance value does **not** meet the specification, replace the resistor and check the resistance value again.

If the harness resistance still does **not** meet specification, replace the harness.



Last Modified: 17-Apr-2014
